

GenAI as an archetype of decomposition, recomposition, and generation

Structural analogies between Generative AI and creative practices that share (at least apparently) its mechanics

Riccardo Bovetti

[Who am I?](#) | [LinkedIn](#) | [All you need is thought](#)

January 2026

Abstract

This is a (rather naive, but honest) attempt to convince myself that GenAI did not show up yesterday to ruin the party, but is mostly formalizing creative tricks humanity has been using for a while—just without necessarily giving them fancy names.

I start from my natural habitat (a serial hoarder of notes, links that die as 404s, and half-sketched ideas) and try to compare the mechanisms of transformers (segmentation, attention, recontextualization, reassembly) with four old artistic acquaintances: Chomskyan generative grammar, collage/cut-up, Eisensteinian montage, and above all musique concrète à la Schaeffer.

The hypothesis is simple: if today we are amazed that a machine can “generate”, it is also because we have conveniently forgotten how many times we have generated (by hand, on tape, with scissors and glue, or with rules) by remixing pre-existing material until something emerged that was not there before—or at least was not visible.

Contents

1	An unrequested introduction	3
2	Introductory synthesis	3
3	Generative grammar: Chomsky’s linguistic algorithm	4
4	Collage and combinatorial writing: from plastic arts to cut-up	5
5	Film montage: Eisenstein’s generative syntax	5
6	Musique concrète as a paradigm of generative process	6
7	Toward a unified theory of cultural generation	7

8	Conclusions	8
	Essential bibliography	10

1 An unrequested introduction

I would like to reassure my readers, including the accidental ones. I am (almost) about to exhaust my archives and therefore—unless I radically change my research approach—my production of negligible cultural content is destined to shrink quite a lot, and fairly soon.

When I say “archives” I mean those immense collections of articles, links (which in most cases lead to 404s), books, notes, sketches, and conceptualizations via more-or-less daring visual metaphors that I have been gathering (compulsively) since university; and that, over the years, I have digitized and collected—for a while in an orderly, structured way, and then progressively in a more and more chaotic and undisciplined way (as the amount of material increased, and the time available to read, select, and classify it decreased accordingly).

I cannot give a single, unambiguous explanation for the reasons behind this behavior: I am an enthusiastic optimist, at least as far as my cognitive ambition (and the time available to satisfy it) is concerned. But whenever I run into something I do not know, if that topic clears even the minimal threshold of my disinterest, I start obsessing over it—not so much by going deep, but by trying to collect as much material as possible, following non-linear paths to find it and connect it.

I try to understand a little—the bare minimum needed to outline the main elements—and then I hoard as much as I can so I will have it available as soon as I have time to study it properly (spoiler: never).

In this, as further proof that the apple never falls very far from the tree, I am not that different from my uncle, who accumulated one or two newspapers for every single day of his 40 years of work, rigorously without ever even skimming one; filling every room of his apartment with boxes of newspapers, with the plan of reading them once he retired.

As the sole evolutionary element of the species, I ended up with hard drives, NAS, CDs, and USB sticks bursting at the seams—but at least I do not run the risk of a spark turning an elderly Piedmontese man into a blazing Hindu on his pyre of knowledge.

The possibilities GenAI has given me (with the techniques I described in the previous article) to dispose of this archive while still extracting value from it—even if not always for the reasons that pushed me to build it in the first place—are remarkable.

Since in these (by now) more than thirty years of serial-hoarding career I have been attracted to topics that are very far apart and, at first glance, totally disconnected from each other, the exercise of finding connections and discovering that, in the end, everything holds together is one of the things I am finding particularly gratifying.

The question remains open, though: what will I do (if/when) I retire?

2 Introductory synthesis

As I move forward in my research and in my attempt to understand how GenAI works, bold parallels (most of the time completely pulled out of thin air) keep coming to mind. They push me to attempt a critical rereading of several pre-existing disciplines, trying to spot and highlight

structural and methodological similarities.

I do not know whether this urgent need is driven by an attempt to soothe my conscience about the objective difficulty of keeping up with these developments—by turning it into a “well, nothing new then . . .” with a slightly sour-grapes flavor—or whether it is simply the fact that I now have tools that let me check, in a very simple way, even my most far-fetched hypotheses.

Either way: in this rather long-winded piece I tried to analyze four different domains of artistic creation that, at different moments in my life, caught my attention and triggered my hoarding instinct: Chomskyan generative grammar, *musique concrète*, artistic collage and experimental combinatorial writing, and experimental film montage. They all share fundamental operating principles with the architecture of transformer algorithms and GenAI attention mechanisms.

Through a systematic comparative analysis (which I chose to focus on Pierre Schaeffer’s *musique concrète* as my privileged theoretical paradigm, because in recent weeks I found myself—again, compulsively—immersing in those sounds), I tried to understand whether strategies of segmentation, decontextualization, selective attention, and reassembly historically precede and conceptually anticipate the computational logics of contemporary Generative AI, and whether they might have influenced them in some way, even implicitly.

Although from the outside GenAI looks like a radical discontinuity compared to everything we have known and used so far, I am convinced that the computational formalization of fundamental creative intuitions that cut across disciplines and epochs can open up new perspectives for a sort of unified theory of generative creativity.

Under the hood of the GenAI systems with which we have now redrawn the boundaries of our everyday digital interaction, there is an engine based on the so-called “attention mechanism”, which uses Query, Key, and Value to create dynamic contextual representations, processing all the tokens (or lexemes) of a sequence simultaneously (i.e., a sentence, a clause).

The core principles that govern these mechanisms—segmentation of pre-existing materials, recontextualization, assembly guided by selective attention dynamics, and the emergence of new meanings—are not exclusive to the contemporary computational domain.

On the contrary, albeit in different forms, they can be found in several consolidated creative and scientific practices, suggesting the existence of universal patterns of recomposition and generation.

3 Generative grammar: Chomsky’s linguistic algorithm

Noam Chomsky’s generative grammar (1950s) is a fundamental theoretical precursor for understanding analogies with GenAI. Chomsky defines language as a finite set of rules capable of recursively generating all grammatical sentences of a language. This formal structure presents surprising parallels with generative neural architectures.

First of all, just as a generative grammar defines explicit syntactic patterns, a large language model implicitly learns co-occurrence patterns and uses them to compose novel texts. In both cases, from a limited set of rules (or learned weights) potentially infinite combinations emerge.

Second, the recursive capacity of Chomskyan grammar mirrors the self-attentive nature of transformers, where each token can refer to any other token in the sequence via self-attention mechanisms. A crucial difference is that, while Chomsky postulated innate rules that generate language, GenAI leverages empirical data (which, however, are themselves derived from natural language). Both, though, raise fundamental questions about the nature of linguistic “understanding” and about the boundary between human and mechanical language production (both in semantic terms and in terms of intellectual property).

4 Collage and combinatorial writing: from plastic arts to cut-up

Collage, developed by Georges Braque and Pablo Picasso in 1912, and subsequent combinatorial writing practices (Dadaist cut-up, *détournement*) are creative techniques that assemble fragments to generate new wholes and semantic tonalities. Picasso used fragments of newspapers, wallpaper, and various materials, recomposing them through layering.

Similarly, Tristan Tzara’s Dada poems (1916) and the experimental works of William Burroughs and Brion Gysin were assemblages of clippings and textual fragments recomposed (pseudo-)randomly. Kenneth Goldsmith describes cut-up as “post-writing” based on the reshuffling of existing materials. Both visual collage and textual cut-up allow elements to develop new meanings through contextual relocation.

This process resembles, in some respects, how GenAI recontextualizes linguistic tokens to generate syntactic emergences to which we sometimes attribute, while reading, unexpected semantics. As an additional point of similarity: *papier collé* integrated textual reality directly into the composition, while cut-up operated through mechanical algorithms of shuffling. Both practices seem to anticipate GenAI’s multi-head attention approach, where different levels of meaning are processed simultaneously in order to select the most relevant one (aesthetically?).

The discussion around these practices, at the time, focused on their role in demystifying the idea of creativity as a purely original gesture. As Goldsmith observes, new meaning is born “solely from the reshuffling of existing materials”, fully aligned with the codes of the era which—as Floridi says—is an era of Design rather than creation or invention, raising questions about the distinction between author and algorithm that precede contemporary debates on artificial intelligence by decades.

5 Film montage: Eisenstein’s generative syntax

Sergei Eisenstein’s theory of experimental montage, in its own way, offers a paradigmatic model of semantic generation through the combination of pre-existing elements. For Eisenstein, montage is not simple chronological concatenation, but a creative operation that produces meanings that cannot be reduced to the individual shots. Shots placed side by side transform one another, generating concepts that are not present, not visible, or not readable on their own within the individual scenes.

As Eisenstein notes, “what results is a movement of transfiguration in which the power of the word survives and takes shape in the image, without becoming its representation.” The film image, understood as an original sequence of frames, retains its own force while contributing to a new semantic context. This dynamic of “survivals” and “mutations” echoes generative image algorithms, which combine visual elements (textures, shapes, colors) to synthesize new ones with emergent meanings through a trial-and-error mechanism involving continuous back-and-forth judging roles between the discriminative and the generative network.

Eisensteinian montage operates like an algorithm that mixes vectors of meaning, suggesting that cinematic creativity can derive from combining mechanical composition processes with the director’s “pure” vision. This intuition anticipates GenAI’s computational approach to generating visual content and also invites comparison with RAG techniques that specialize (by vectorizing it) specific knowledge and meaning.

6 Musique concrète as a paradigm of generative process

Musique concrète was born in 1948 from the meeting of the technical needs of broadcasting and Pierre Schaeffer’s aesthetic intuition at the RTF in Paris. This historical moment took place in an avant-garde climate that had already seen the Italian Futurists (Luigi Russolo, “L’arte dei rumori” 1913) theorize the musicality of industrial noises, and John Cage anticipate the use of unconventional sounds in place of, instead of, and in combination with musical notes.

Musique concrète enacts a fundamental break with the Western epistemological tradition: instead of starting from an abstract musical idea to notate and hand to instruments via a score written *a priori*, one collects concrete sounds and abstracts the musical values they potentially contain. This methodological inversion—from the concrete to the abstract—anticipates the data-driven approach of all analytical mechanisms (powered by probabilistic and statistical engines) that underpin today’s version of AI.

In the case of musique concrète, the use of magnetic tape is not merely technical and functional but constitutive and existential: it allows the decontextualization of sound, its fixation onto a medium, and therefore its manipulation separate from the original context in which it was produced. This “reification” of sound anticipates (let us take the concept with a pinch of salt) the vector representation of embeddings in GenAI.

Schaeffer introduces the revolutionary concept of the “sound object”—sound contemplated in its full phenomenological completeness (attack, timbre, duration, envelope, frequency) rather than as a vehicle for abstract harmonies or melodies. This object, “torn from the temporality of performance” through the circularity of vinyl, becomes an autonomous, manipulable unit—a purely syntactic structure onto which a semantic layer can be draped, depending (to a large extent) on the listener’s knowledge and sensitivity.

Acousmatic listening (hearing sound without seeing its source), which is part of the concrète experience, makes it possible to focus precisely on the syntactic qualities of sound, freeing them from causal associations or overly rigid semantics. This process of guided abstraction resonates

with GenAI mechanisms aimed at extracting abstract patterns from training data, making manipulable representations of content described by a shared syntax.

Pierre Henry defines *musique concrète* as focused on “envelopes and forms” rather than timbres, aiming to “plasticize” music like sculpture. This materialization of sound foreshadows the continuous manipulability of neural representations and, in particular, recalls the “for mere mortals” descriptions of convolutional networks.

As with the other techniques introduced above, the most interesting part—the one from which analogies can be extracted and appreciated—is the “creative” process. In *musique concrète*, the compositional protocol proceeds through recording real sounds, tape manipulation (slowing down, speeding up, cutting, re-splicing, filtering, layering), and final assembly. A train sound becomes a pulsing rhythm if sped up; a voice becomes alien if slowed down.

The *concrète* composer functions as a “human algorithm”, or more precisely as the executor of an algorithm, applying technical parameters (speed, filter, intensity) to the input sound in order to obtain musical output. Unlike “traditional” music, one does not start from a written score but from “raw sound”, eliminating notational mediation. As Schaeffer notes, *concrète* “does not rely on preliminary notations” and does not follow pre-established harmonic schemes. The composer-researcher explores, through tape manipulation, the “forms” that emerge from sonic combinations: this process is analogous and anticipatory (logically speaking) to exploring latent space in multimodal GenAI.

Like other forms of “liminal” art, *concrète* challenges traditional aesthetic categories, treating any sonic material as potentially musical. This democratization of sonic material foreshadows the theoretical approach of training LLMs, in which (at least at the beginning) all lexemes must be treated as having equal semantic potential (initial model weights, to put it slightly more precisely).

Musique concrète teaches that musical creativity can derive from recombining existing materials. When Schaeffer composes with other people’s recordings, questions of authorship emerge that are analogous to contemporary ones: how much “originality” does a composer have if they use only pre-existing sounds? To what extent is creative work detached from the author and anchored to the given material? Multimodal GenAI pushes the questions raised by *concrète* to an extreme, proposing a “music without musicians” that, like *concrète*, radically redefines what we mean by composition and musical listening.

7 Toward a unified theory of cultural generation

The comparative analysis reveals the existence of meta-disciplinary principles of generation that operate across apparently heterogeneous domains. These principles constitute the foundations of a possible unified theory of generative creativity, which could be articulated through four foundational principles:

1. **Productive decontextualization:** the controlled removal from the original context as a

necessary precondition for new semantic emergences.

2. **Distributed attention:** selective weighting mechanisms allow dynamic focus while maintaining awareness of the whole.
3. **Emergent recombination:** assembly strategies that generate qualitative properties irreducible to the components taken individually.
4. **Generative recursion:** the ability to apply the same transformation operations recursively, producing a formal fabric of any length.

GenAI, whether considered in its operating dynamics or in its application to artistic forms, does not represent a radical discontinuity in the history of creative practices, but rather the computational formalization and exponential acceleration of pre-existing generative strategies.

This perspective suggests three fundamental epistemological considerations:

- By conducting a retrospective analysis starting from GenAI’s operating mechanisms, it is possible to recognize operational patterns in historical practices that had not been theorized in these terms.
- While the mechanisms are historically continuous, GenAI introduces a quantitative discontinuity (speed, scale, precision) that, through sheer repetition, produces qualitatively new effects.
- GenAI makes accessible—unfortunately?—at mass scale, creative processes previously limited to specialized artistic or scientific elites, raising new questions about the nature of creativity and cultural economics.

This epistemological reassessment suggests continuing the systematic analysis of consolidated artistic and scientific practices to identify computational principles implementable in advanced neural architectures. On the other hand, integrating the results of the analysis of creative domains will certainly help develop even more sophisticated neural architectures. As already mentioned in my previous article, it is also desirable to use advanced GenAI techniques to reanalyze (primarily literary) historical corpora according to novel interpretive paradigms, creating new hermeneutic perspectives.

8 Conclusions

With hindsight—which makes everyone a master—GenAI looks less and less like a technological rupture, and more like the computational systematization of fundamental creative intuitions that cut across disciplines and epochs. This historical continuity does not diminish the importance of the contemporary revolution, but inscribes it into a broader cultural genealogy that helps us understand its meaning and potential. Creativity and computation are not antithetical but intimately connected: every creative act implies forms of computation (selection, transformation,

combination), while every advanced computational process manifests emergent creative properties. Understanding GenAI through this historical-comparative lens not only enriches our interpretation of the technological present, but also suggests that the future of artificial intelligence could benefit significantly from dialogue with the history of human creative practices.

Essential bibliography

References

Generative AI and neural architectures

- [1] Vaswani, A., et al. (2017). Attention is all you need. *Advances in neural information processing systems*, 30.

Linguistics and generative grammar

- [2] Chomsky, N. (1957). *Syntactic Structures*. Mouton de Gruyter.
- [3] Chomsky, N. (1965). *Aspects of the Theory of Syntax*. MIT Press.

Musique concrète

- [4] Schaeffer, P. (1952). *À la recherche d'une musique concrète*. Éditions du Seuil.
- [5] Schaeffer, P. (1966). *Traité des objets musicaux*. Le Seuil.
- [6] Russolo, L. (1913). *L'arte dei rumori*. Futurist manifesto.

Art and collage

- [7] Goldsmith, K. (2011). *Uncreative Writing: Managing Language in the Digital Age*. Columbia University Press.

Experimental literature and cut-up

- [8] Goldsmith, K. (2015). *Capital: New York Capital of the 20th Century*. Verso Books.

Cinema and montage

- [9] Eisenstein, S. M. (1949). *Film Form: Essays in Film Theory*. Harcourt Brace.
- [10] Eisenstein, S. M. (1942). *The Film Sense*. Harcourt Brace.